

Geotechnical Application for Expanded Polystyrene Waste

Published 2014 • Environmental Science, Geology, Engineering

Expanded polystyrene (EPS) packaging consumes landfill space when discarded. Reusing this waste product would relieve landfill pressures and provide geotechnical benefits. Previous studies of waste materials for soil reinforcement highlighted the potential for EPS waste to be used similarly. EPS beads of varying diameters and concentrations were tested for their influence on friction angles and dry density of sandy soils. Results showed decreases in the friction angle under heavier loads with... [Expand](#)

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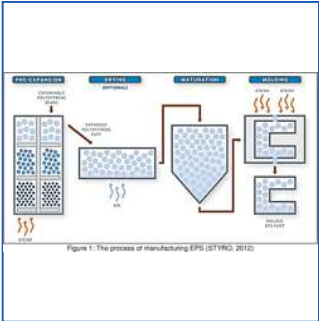


Figure 1

PHYSICAL PROPERTIES		
Property	Unit	Value
Specific Gravity	g/cm ³	1.03-1.06
Apparent Density	g/cm ³	0.6-0.65
Water Absorption	%	0.03-0.1
MECHANICAL PROPERTIES		
Young's Modulus	MPa	3000-3600
Tensile Strength	MPa	30-60
Tensile Elongation	%	1.0-5.0
Shear Modulus	MPa	1400
Flexural Modulus	MPa	3200

Table 1



Figure 2

Characteristics	Unit	Cape Flats Sand	Whepswood Sand
Specific Gravity, G _s	Mg/m ³	2.65	2.65
Natural Moisture Content	%	3.09	3.65
Average Densest Dry Density	kg/m ³	1728	1801
Average Loose Dry Density	kg/m ³	1620	1692
Optimum Moisture Content (Proctor)	%	13.9	7.8
Maximum Dry Density (Proctor)	kg/m ³	1801	1805
Particle Range	mm	0.060-1.18	0.060-2.00
Mean Grain Size, D ₅₀	mm	0.29	0.24
Coefficient of Uniformity, C _u	-	2.07	2.15
Coefficient of Curvature, C _c	-	1.04	1.10
Angle of Friction, φ (Peak)	degrees	40.5	39.0
Apparent Cohesion, c' (Peak)	kN/m ²	0	5.0

Table 2

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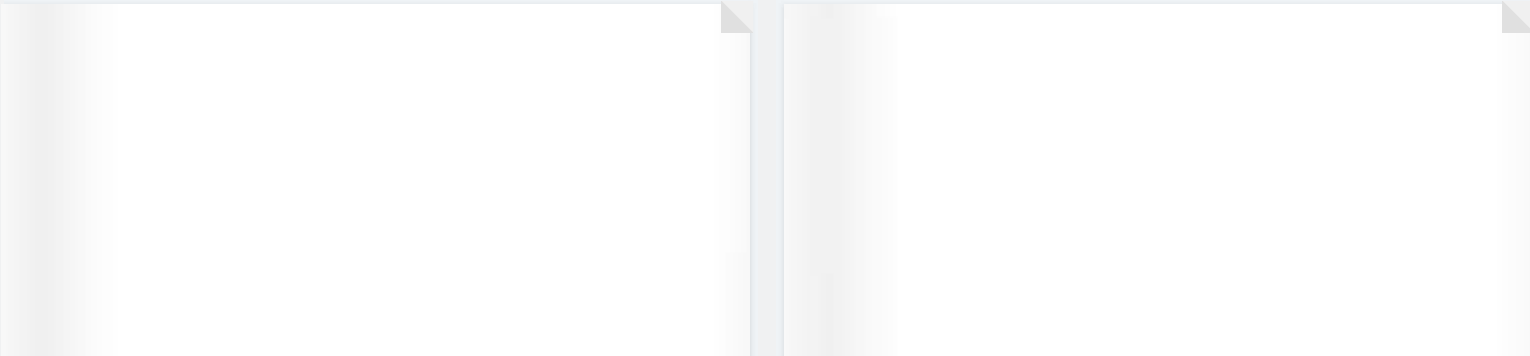
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